



Luck is what happens when preparation meets opportunity.

Agenda

Class starts @ 9:03 pm

- ① Databricks spark demo
- ② Delta lake and Delta Table
- ③ Delta Table demo

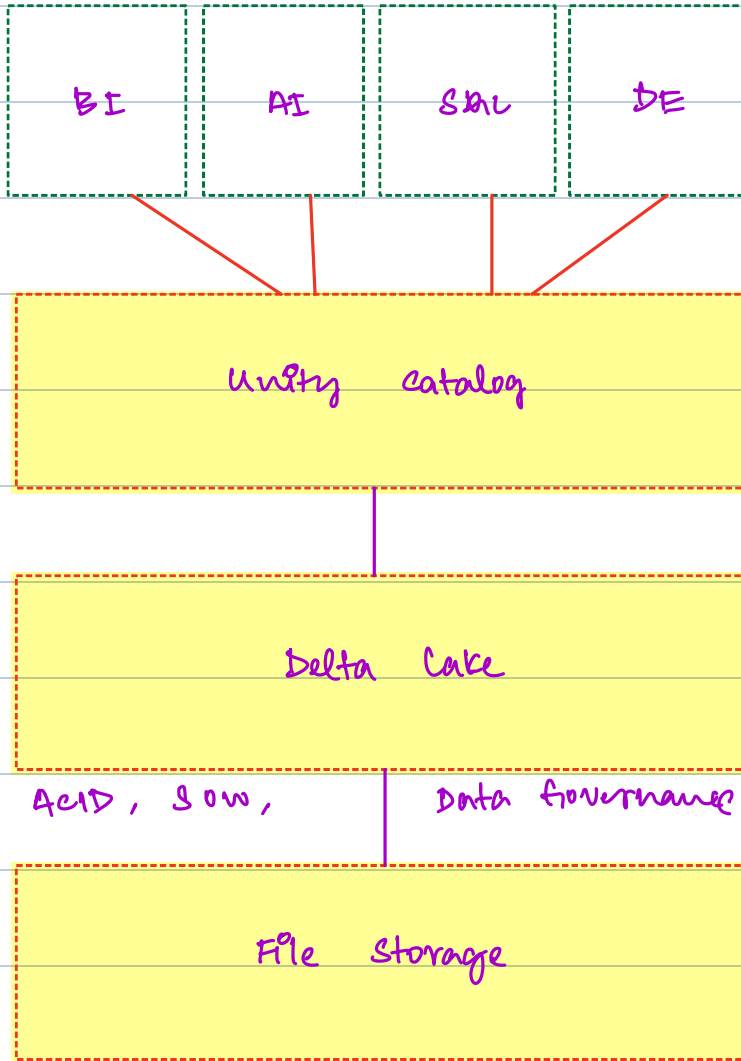
Databricks

-
- ① Snowflake introduction
 - ② Snowflake architecture
 - ③ Components of Snowflake

Snowflake

- ④ Hands-on snowflake
- ⑤ Snowflake vs BigQuery vs Amazon Redshift vs Databricks

Databricks



1980 - Structured data
earn profit

DWH (structured)
Infra + compute

2000's (semi structured/
non-structured)

Data Lake

Implemented
using

DELTA TABLE



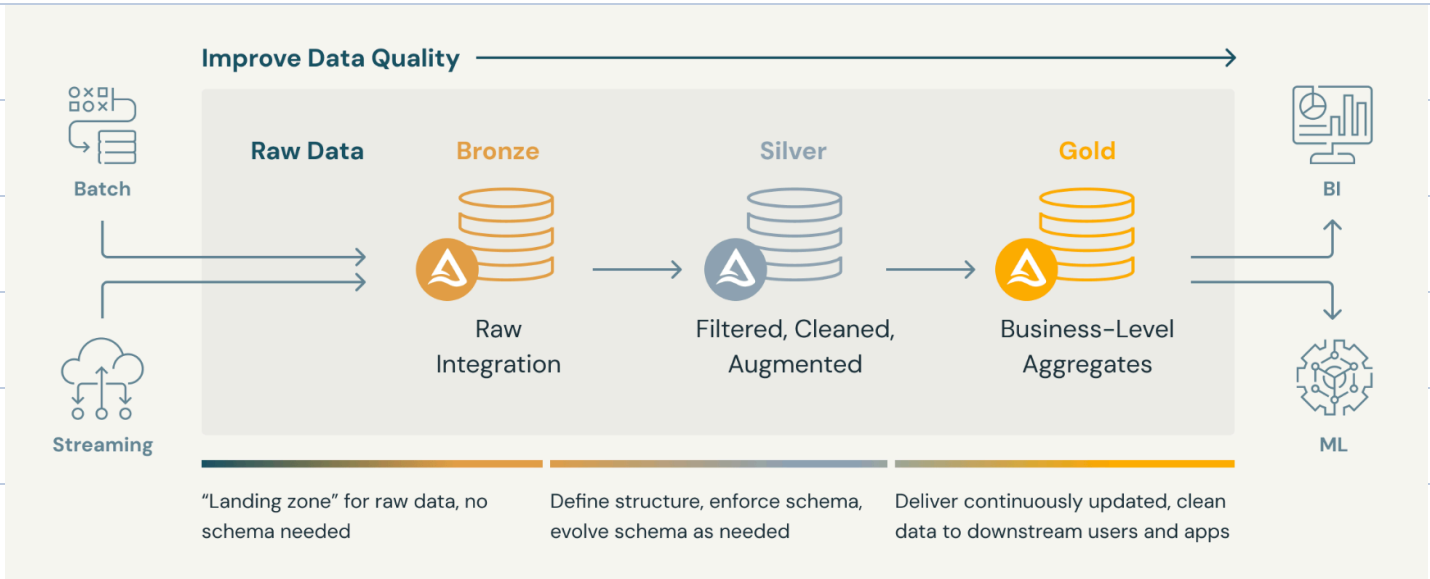
- ACID Transactions
- Time Travel
- Schema Enforcement & Evolution
- Efficient Updates & Deletes
- Streaming + Batch
- Performance Boost

PARQUET/CSV



- No ACID Compliance
- No Versioning
- Schema Enforcement Only
- Append-Only
- Separate Batch & Streaming
- Limited Performance

Medallion Architecture



Categorises data based on its quality.

Bronze:

Data in its raw format.

- ① check for schema
- ② check for nulls
- ③ check for mistyped data

Silver: filtered and cleaned data

Golden: Business level aggregations

[all business logic for a given use case]

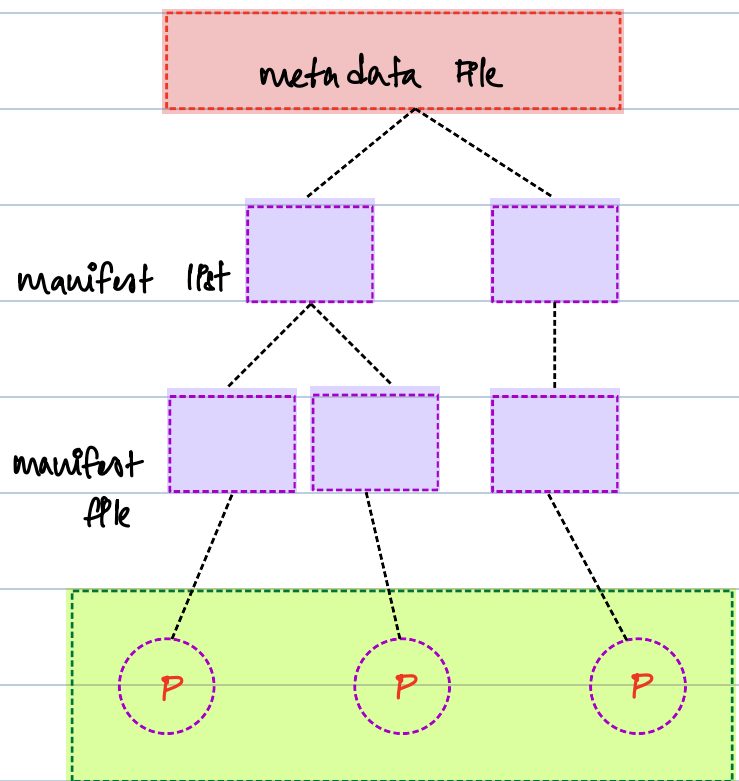
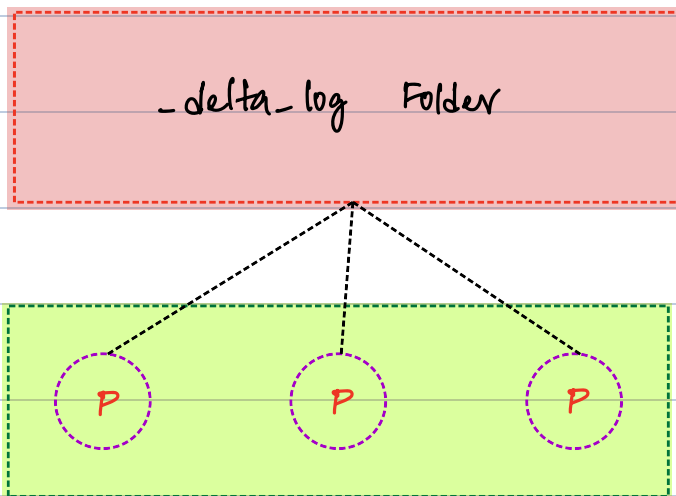
Delta lake vs Iceberg

```
my_delta_table/
├── _delta_log/
│   ├── 00000000000000000000000000000000.json
│   ├── 00000000000000000000000000000001.json
│   ├── 00000000000000000000000000000002.json
│   ├── ... (more json commits as transactions occur)
│   └── 00000000000000000000000000000010.checkpoint.parquet
├── part-00000-a1b2c3d4-e5f6-g7h8-i9j0k1l2m3n4.c000.snappy.parquet
├── part-00001-b5c6d7e8-f9g0-h1i2-j3k4l5m6n7o8.c000.snappy.parquet
└── part-00002-c9d0e1f2-g3h4-i5j6-k7l8m9n0o1p2.c000.snappy.parquet
```

```
my_iceberg_table/
├── data/
│   ├── part-00000-a1b2c3d4-e5f6-g7h8-i9j0k1l2m3n4.c000.snappy.parquet
│   ├── part-00001-b5c6d7e8-f9g0-h1i2-j3k4l5m6n7o8.c000.snappy.parquet
│   └── ... (more data files are added here)
└── metadata/
    ├── 111-abc-manifest.avro
    ├── 222-def-manifest.avro
    ├── ... (more manifest files)
    ├── snap-1111-1-abc.avro
    ├── snap-2222-1-def.avro
    ├── ... (more manifest lists, one for each snapshot)
    ├── v1.metadata.json
    ├── v2.metadata.json
    ├── v3.metadata.json
    └── version-hint.text
```

+ metadata on top of your original Data

ACID, Schema on write,
Data History (- delta-log)

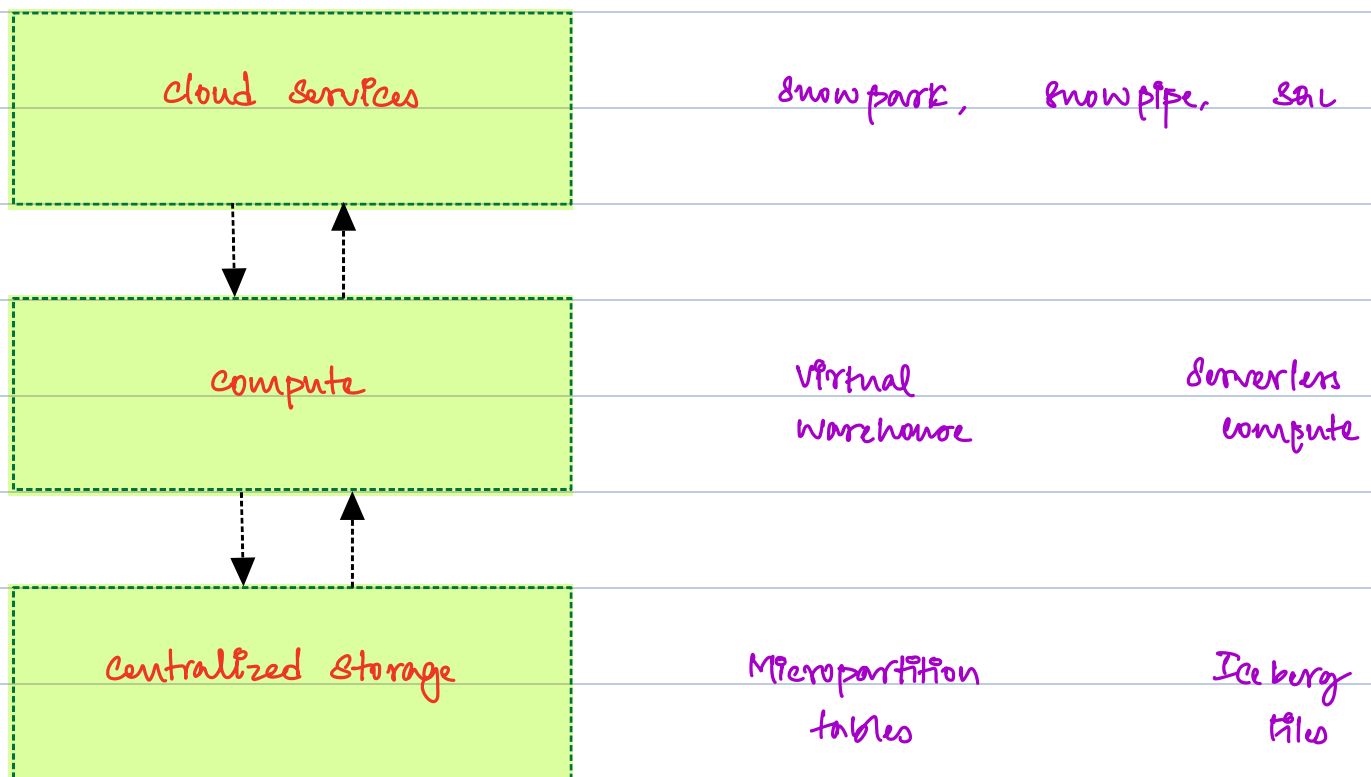


product → category = "mobile"
id = 123

Snowflake

Snowflake is a cloud native data platform, provides a single unified system for a vast range of data workloads.

- ① Separates storage from compute resources
- ② Data cloud that enables Data Warehouse, Data Lakehouse, Data engineering, Data science .. etc.



Architectural Pillars and Components

- ① SQL API: used for interacting with SQL in cloud services [DW, HL]

Different components in snowflake
examples of DWH

Spark streaming